



Philly Cheesesteaks

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Introduction



Motivation:

The restaurant industry is highly competitive and difficult to succeed in. This project asks whether data-driven insights from reviews and business features can help identify what truly drives restaurant ratings.

Our approach:

Customer ratings strongly influence restaurant success, but reviews mix emotional reactions with structural business factors. This project uses modeling to separate these signals and identify data-backed factors that drive restaurant ratings.

Goal

- To determine what factors are most strongly associated with higher Yelp review ratings, specifically for Philly Cheesesteak restaurants
- Provide actionable recommendations to business stakeholders



Data Insights

We found:

- Customer satisfaction is the **strongest** predictor of higher Yelp stars
- Other features are much weaker in comparison

Why this matters:

- Data-driven insights from Yelp translate to well-known conclusions
- There are no shortcuts for restaurant success; the customer is of greatest importance!

Methodology

- Used* businesses.csv and reviews.csv
- Extracted relevant features from noisy data
 - Evaluated average review sentiment* for each business
 - One-hot-encoded attributes like “DietaryRestrictions.halal”, “BYOB”, “Open24Hours”, etc.
- Imputed missing values with ‘0’ - cannot guarantee missing at random
- Filtered for only “Cheesesteak” restaurants
- Standardized scaling for interpretable model coefficients
- Fit linear regression to with target variable = Yelp “stars”

*summary of data available in appendix (1)

*more on sentiment analysis in appendix (5)

Model

About:

- Linear regression (LassoCV)
- 5-fold outer CV, 5-fold inner CV for regularizing

Why Lasso?

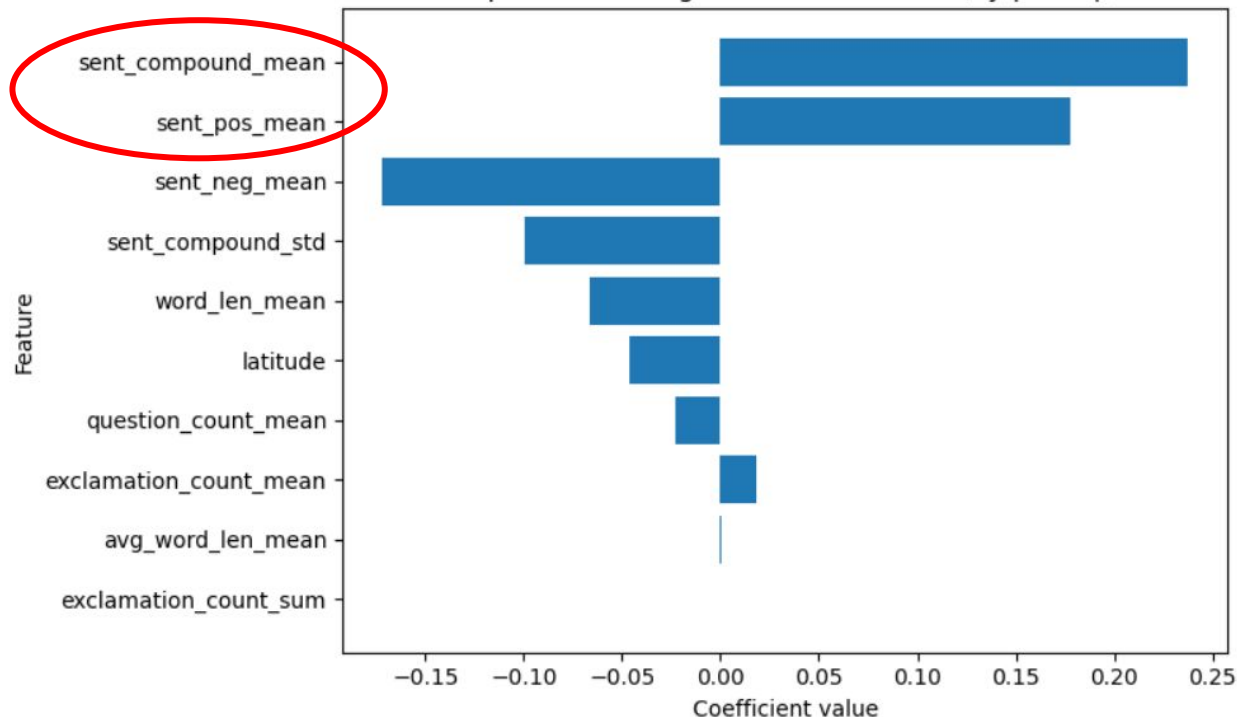
- Commonly used in biological sciences to identify influential features

Metrics:

- CV RMSE: 0.4045 ± 0.0193
- CV R^2 : 0.7083 ± 0.0423
- Model is quite accurate for a linear regression on noisy data

Important Model Features

Top 10 Linear Regression Coefficients (by |value|)

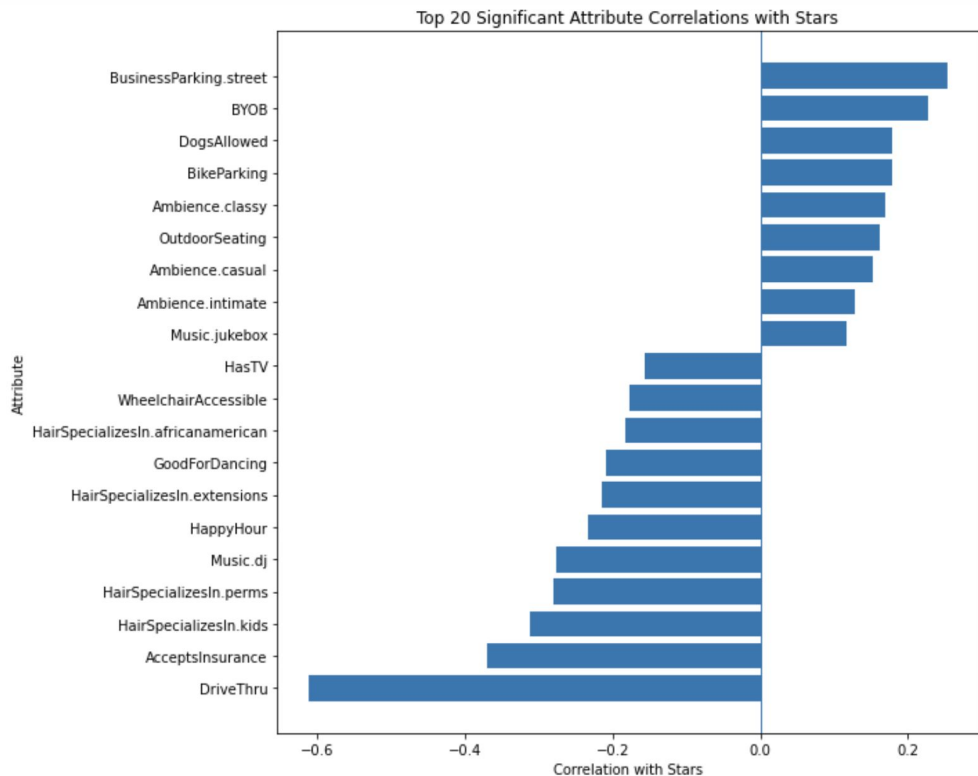


- Sent_compound_mean = average positivity score of reviews
- Sent_pos_mean = average proportion of positive words in review

Interpretation: **customer satisfaction is the strongest predictor**

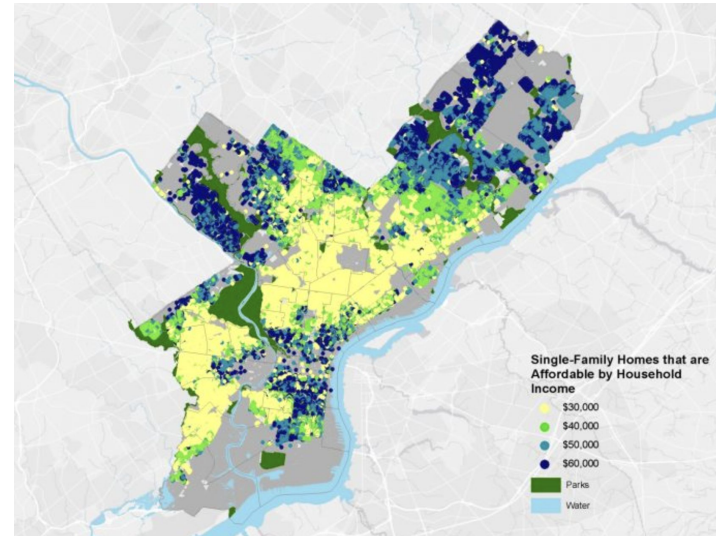
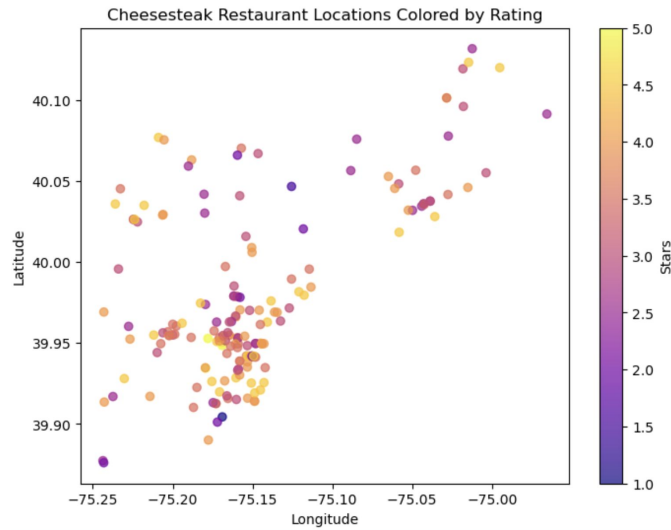
*all coefficients available in appendix (2)

Important Non-text Features



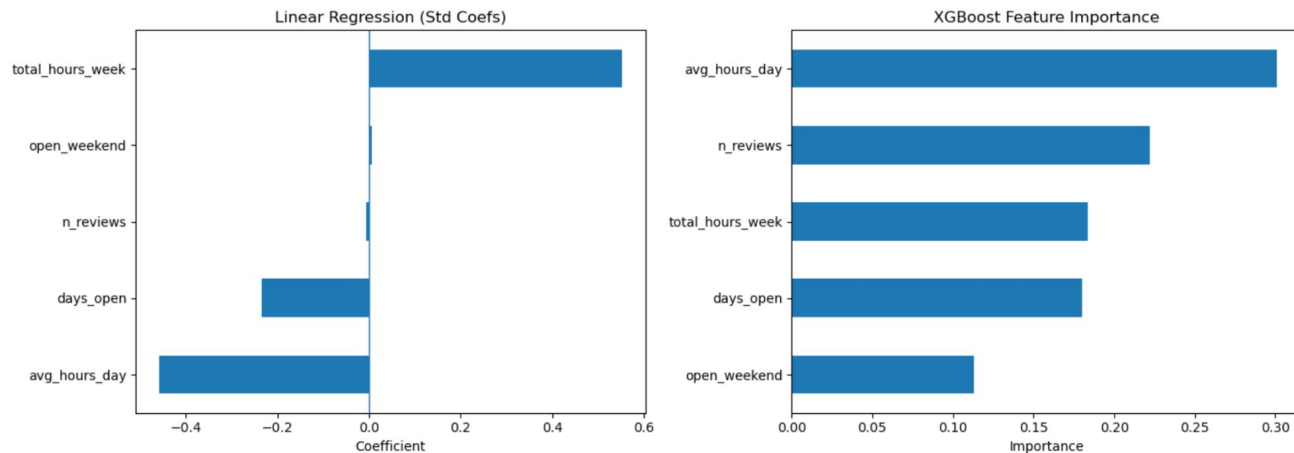
- All non-text features have correlations with strength less than 0.3, with a few exceptions
- Customer satisfaction-related features dominate other information like TV, Happy Hour, etc.

*all correlations available in appendix (3)



<https://econsultsolutions.com/mapping-the-affordability-in-philadelphia/>

Store Opened Time Period



*appendix 4

Actionable Recommendations:

How can we have better reviews?

- **Prioritize better customer service**

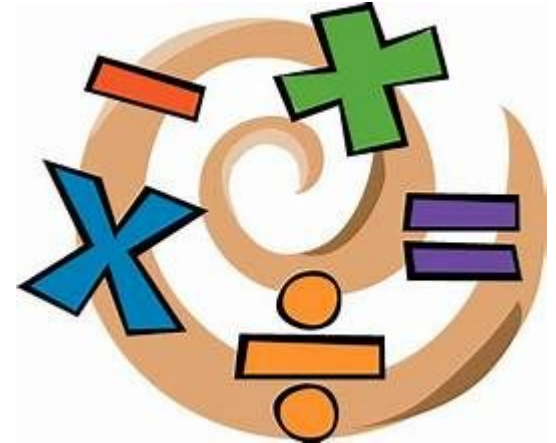
Over time:

- Focus on maintaining positive experiences and keep a strong Yelp image



Appendix

1. Data summary statistics
2. Model feature coefficients
3. Correlations of non-text features
4. Correlation between Store Opened Time Period and Customer Review
5. Sentiment Analysis



1) Data Summary Statistics

- Number of rows in businesses.csv: 14569
- Number of rows in reviews.csv: 967552
- Columns with highest percentage of missing values in businesses.csv: hours (19%), attributes (8%)
- Columns with lowest percentage of missing values in businesses.csv: postal_code (5%)

Review-level stars summary:

- Mean: 3.786
- Median: 4.000
- Variance: 1.927



2) Model feature coefficients

sent_compound_mean	0.2373420101840284
sent_pos_mean	0.17777421717134595
sent_neg_mean	-0.171714893
sent_compound_std	-0.099500946
word_len_mean	-0.066487605
latitude	-0.046291398
question_count_mean	-0.022622775
exclamation_count_mean	0.018256229
avg_word_len_mean	0.000337506
exclamation_count_sum	0
word_len_std	0
question_count_sum	0
char_len_std	0

Sorted by absolute importance, decreasing

The following are all 0 importance:

exclamation_count_sum, word_len_std, question_count_sum, char_len_std, char_len_mean, sent_compound_max, sent_compound_min, caps_ratio_mean, postal_code, sent_neg_std, sent_neu_mean, sent_pos_std, is_cheesesteak, BestNights, Music, HairSpecializesIn, Ambience, GoodForMeal, BusinessParking, RestaurantsPriceRange2, is_open, review_count, longitude, business_id_count

3) Correlations of non-text features

variable	corr	p_value	N
BusinessParking.street	0.25297834508466227	2.8434948202827385e-133	9128
BYOB	0.2276474196868049	4.897927577064434e-07	478
DogsAllowed	0.17905295011047767	7.136683905705229e-14	1722
BikeParking	0.17901837086898703	3.92752817934805e-53	7230
Ambience.classy	0.1688264931299477	4.485601826870674e-30	4492
OutdoorSeating	0.1627652800753082	8.288884628298163e-33	5303
Ambience.casual	0.1528149748538389	3.103971219811057e-25	4561
Ambience.intimate	0.12785842767687883	1.5887219704206597e-17	4407
Music.jukebox	0.11682149504802308	0.001991836	698
Ambience.trendy	0.10936373804325278	5.192046367306651e-13	4335
GoodForMeal.dinner	0.10359409997812963	2.647117870000686e-08	2871
Ambience.hipster	0.10226754466669184	1.174511103751408e-11	4379
RestaurantsReservations	0.099933636	1.5620398185035777e-12	4981
GoodForMeal.brunch	0.097268691	5.002552233851227e-07	2660
ByAppointmentOnly	0.094940092	5.7080961662191165e-09	3750
Caters	0.090067954	3.792223096321684e-09	4266
RestaurantsTableService	0.084356881	0.00012223	2069
GoodForMeal.lunch	0.083293333	1.601420497759625e-05	2676
Ambience.romantic	0.073875275	7.089327010138202e-07	4496
Corkage	0.061367846879789856	0.146960771	560
BestNights.sunday	0.053883008	0.17069171851970827	648
GoodForMeal.breakfast	0.046868712	0.013457288613051906	2780

Sorted by decreasing correlation,
from + down to -

3) Correlations of non-text features (cont)

GoodForMeal.dessert	0.045812628	0.017282517270177815	2700
BestNights.tuesday	0.043604546	0.26770004355958743	648
BestNights.wednesday	0.042089171	0.2846999488698348	648
is_open	0.040414334	1.062632945528387e-06	14569
BusinessParking.garage	0.038355676	0.000263402	9046
BestNights.friday	0.030750259	0.43453946576901686	648
HairSpecializesIn.asian	0.030545466	0.7415726149691515	119
RestaurantsTakeOut	0.026547648612547596	0.036198476	6226
Ambience.upscale	0.024212389	0.10307397079253257	4534
BusinessAcceptsBitcoin	0.023828271	0.3067216361096337	1842
BusinessParking.validated	0.023263086551190786	0.026204092358756308	9133
GoodForKids	0.020855182	0.11335021108186266	5765
BestNights.thursday	0.018206251021443846	0.6436518076888376	648
RestaurantsGoodForGroups	0.011439714722458834	0.4204426101544158	4962
Music.background_music	0.009677201	0.7862163742486663	788
HairSpecializesIn.coloring	0.0040638	0.9650135493487583	119
Ambience.divey	0.002373093	0.8753922107659683	4369
Music.karaoke	-9.01E-05	0.9981408535689854	672
BestNights.monday	-0.003314552	0.9328878452161569	648
BusinessParking.valet	-0.009130056	0.37717555684023557	9358
Music.live	-0.012545476	0.7303839936885932	757
Ambience.touristy	-0.019894471	0.18180673160680663	4506
CoatCheck	-0.022665785	0.5981818782734972	543

BestNights.saturday	-0.038098055	0.3328959349976146	648
Music.video	-0.049175124	0.1676075520594342	789
RestaurantsDelivery	-0.062326788	2.4479125409491225e-06	5708
BusinessAcceptsCreditCards	-0.06852128	5.524901923051075e-14	12015
GoodForMeal.latenight	-0.070969744	0.000266451	2635
HairSpecializesIn.straightperms	-0.082622384	0.37168643897592185	119
BusinessParking.lot	-0.107297446	4.247066588957673e-25	9251
Open24Hours	-0.124034735	0.769809147	8
HairSpecializesIn.curly	-0.140706921	0.12692676577823464	119
HasTV	-0.157268298	1.500242984001512e-28	4908
WheelchairAccessible	-0.176484504	2.2293705414157935e-20	2707
HairSpecializesIn.africanamerican	-0.183522948	0.045730761921079964	119
GoodForDancing	-0.208932295	1.1925233807673548e-06	531
HairSpecializesIn.extensions	-0.213771549	0.019574224	119
HappyHour	-0.233625654	3.7692606578349977e-22	1671
Music.dj	-0.276459726	9.5361761899263e-15	757
HairSpecializesIn.perms	-0.279810151	0.00205645	119
HairSpecializesIn.kids	-0.311060877	0.000574682	119
AcceptsInsurance	-0.369370775	6.272646918976446e-16	448
DriveThru	-0.611058808	8.436100693783012e-42	395

4) Correlation between Store Opened Time Period and Customer Review

	Linear (std coef)	XGBoost (importance)
total_hours_week	0.552491	0.183775
avg_hours_day	-0.458775	0.301012
days_open	-0.234412	0.180253
n_reviews	-0.006634	0.221823
open_weekend	0.006488	0.113136

5) Sentiment Analysis explained

We used a premade `SentimentIntensityAnalyzer` object from the `nltk` library.

VADER is a simple tool within `nltk` that measures sentiment in text.

It looks at the words people use in a review and estimates whether the overall tone is positive, negative, or neutral. It also pays attention to things like emphasis (exclamation points, capitalization) and negation (“not good”).

VADER is especially good for online reviews and social media text, and it works without any training data.